

Fragile Watermarking: Visual Evaluation Report

Generated: May 04, 2026 18:24 | Total Experiments: 32

Experiment 1: Peppers — Attack: Crop 33%



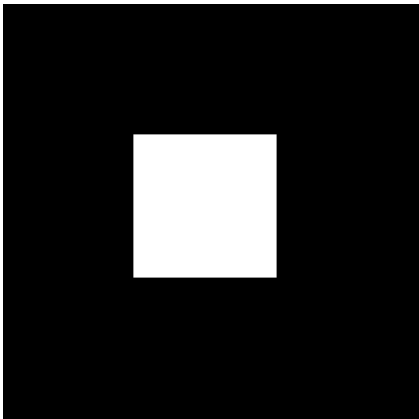
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	32.38 dB
Watermarked SSIM	0.9972	Recovered SSIM	0.9562
True Positive Rate (TPR)	1.0000	Recovered NC	0.9989
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 2: Peppers — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	36.85 dB
Watermarked SSIM	0.9972	Recovered SSIM	0.9814
True Positive Rate (TPR)	1.0000	Recovered NC	0.9996
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 3: Peppers — Attack: Salt_and_Pepper_1%



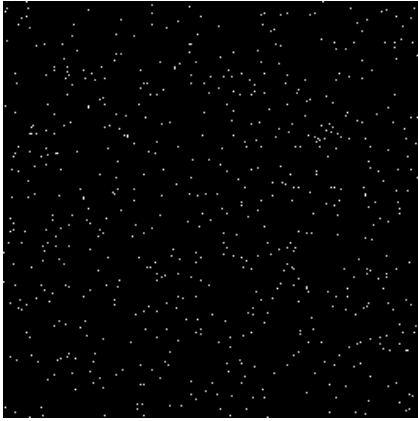
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	28.98 dB
Watermarked SSIM	0.9972	Recovered SSIM	0.8864
True Positive Rate (TPR)	0.5760	Recovered NC	0.9976
False Positive Rate (FPR)	0.0433	Tamper F1-Score	0.1897

Experiment 4: Peppers — Attack: JPEG_80



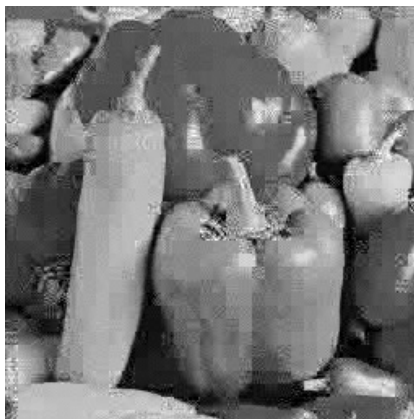
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	24.96 dB
Watermarked SSIM	0.9972	Recovered SSIM	0.7197
True Positive Rate (TPR)	1.0000	Recovered NC	0.9940
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

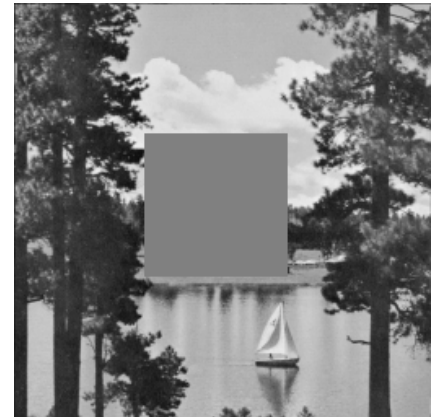
Experiment 5: Sailboat on lake — Attack: Crop 33%



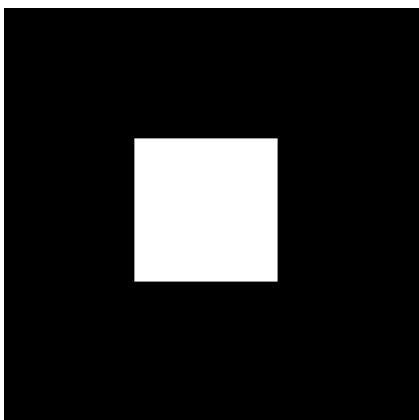
1. Original



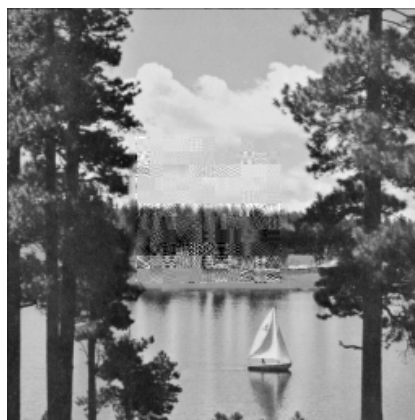
2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	30.97 dB
Watermarked SSIM	0.9976	Recovered SSIM	0.9454
True Positive Rate (TPR)	1.0000	Recovered NC	0.9987
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 6: Sailboat on lake — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



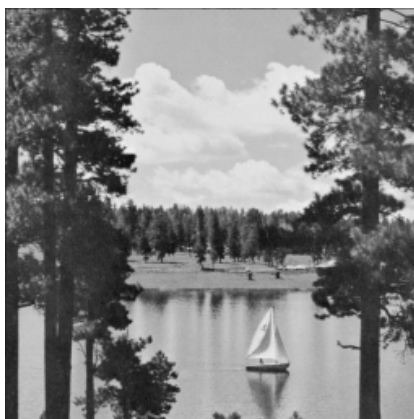
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	35.39 dB
Watermarked SSIM	0.9976	Recovered SSIM	0.9785
True Positive Rate (TPR)	1.0000	Recovered NC	0.9995
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

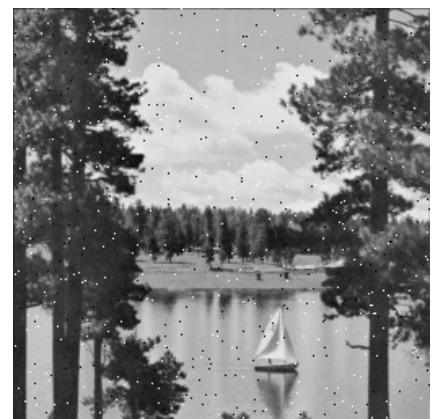
Experiment 7: Sailboat on lake — Attack: Salt_and_Pepper_1%



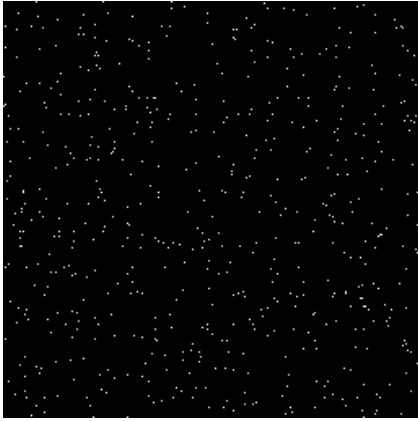
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	27.27 dB
Watermarked SSIM	0.9976	Recovered SSIM	0.8899
True Positive Rate (TPR)	0.5721	Recovered NC	0.9969
False Positive Rate (FPR)	0.0405	Tamper F1-Score	0.1912

Experiment 8: Sailboat on lake — Attack: JPEG_80



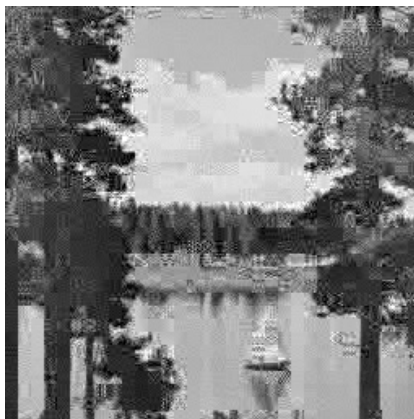
1. Original



2. Watermarked



3. Attacked

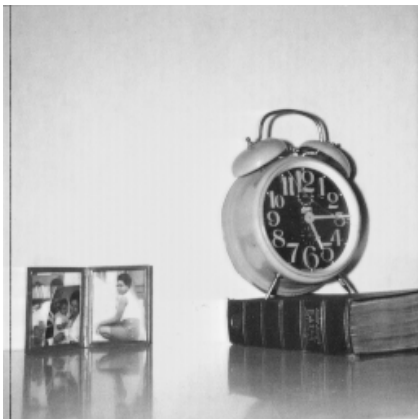


4. True Tamper Map

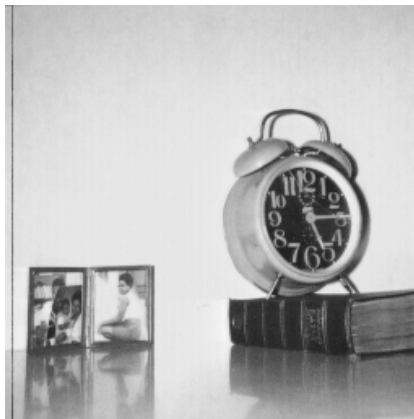
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	21.24 dB
Watermarked SSIM	0.9976	Recovered SSIM	0.6069
True Positive Rate (TPR)	1.0000	Recovered NC	0.9877
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

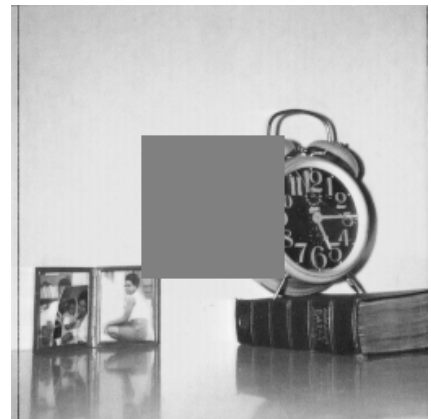
Experiment 9: Clock — Attack: Crop 33%



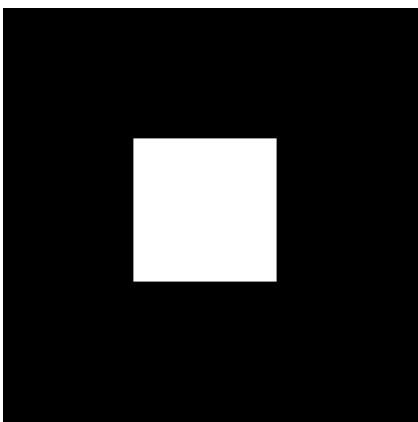
1. Original



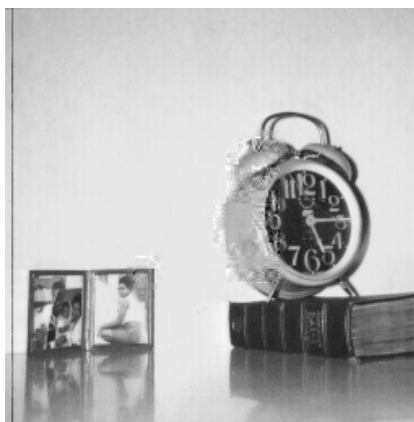
2. Watermarked



3. Attacked



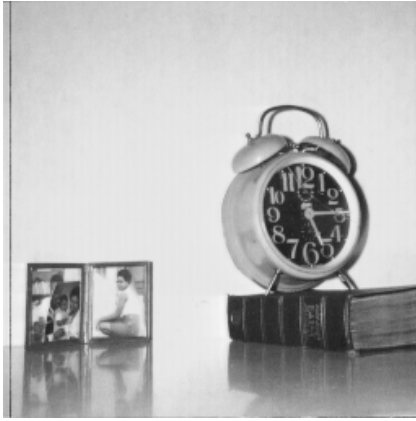
4. True Tamper Map



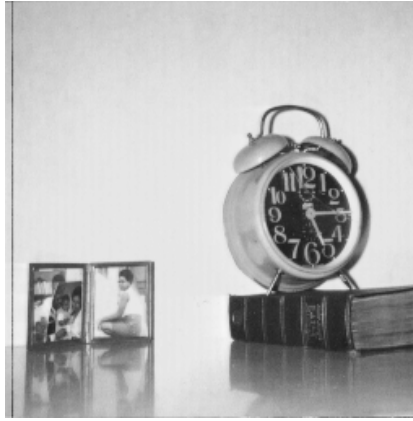
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	34.40 dB
Watermarked SSIM	0.9951	Recovered SSIM	0.9724
True Positive Rate (TPR)	1.0000	Recovered NC	0.9997
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

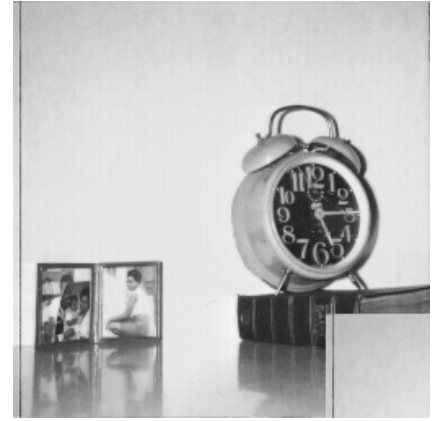
Experiment 10: Clock — Attack: Copy-Move



1. Original



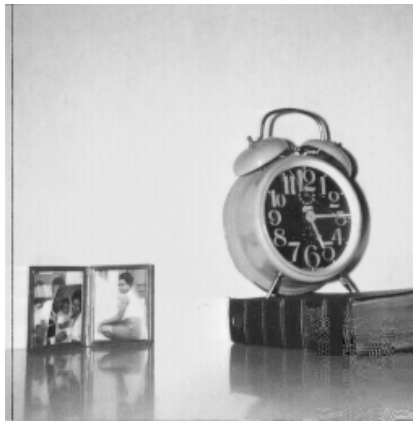
2. Watermarked



3. Attacked



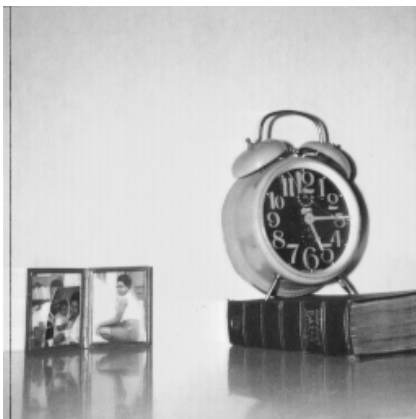
4. True Tamper Map



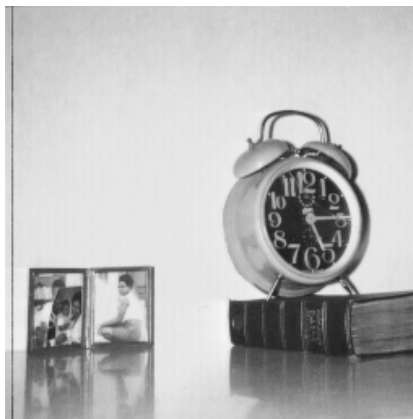
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	37.77 dB
Watermarked SSIM	0.9951	Recovered SSIM	0.9734
True Positive Rate (TPR)	1.0000	Recovered NC	0.9999
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 11: Clock — Attack: Salt_and_Pepper_1%



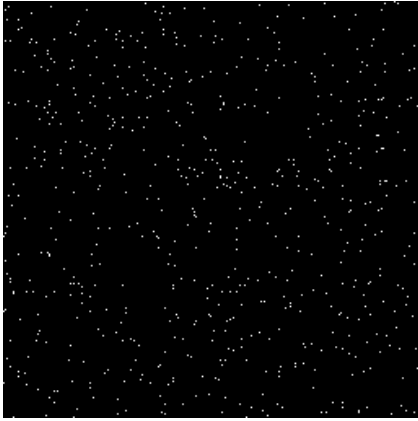
1. Original



2. Watermarked



3. Attacked



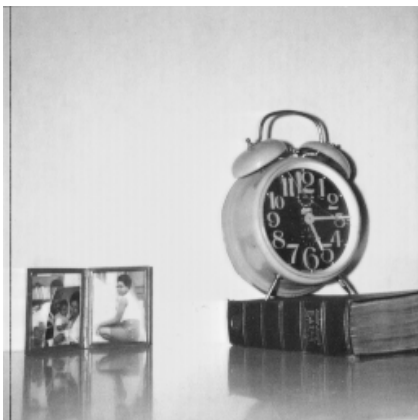
4. True Tamper Map



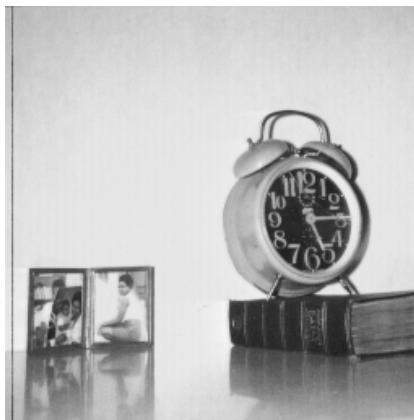
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	28.06 dB
Watermarked SSIM	0.9951	Recovered SSIM	0.8935
True Positive Rate (TPR)	0.5833	Recovered NC	0.9987
False Positive Rate (FPR)	0.0439	Tamper F1-Score	0.1924

Experiment 12: Clock — Attack: JPEG_80



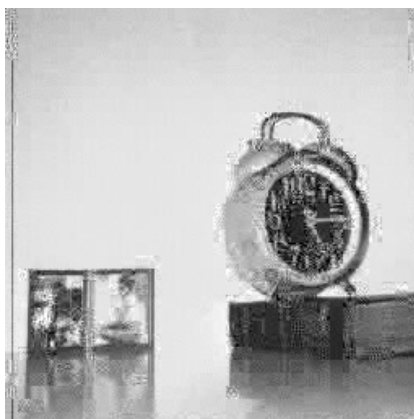
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

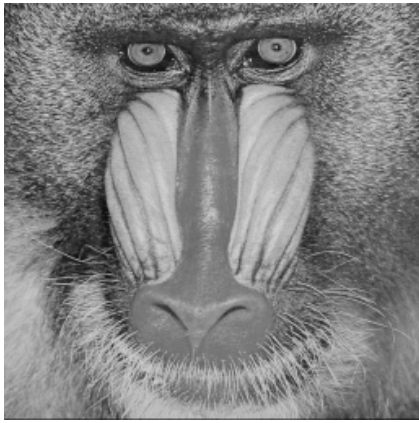
Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	25.71 dB
Watermarked SSIM	0.9951	Recovered SSIM	0.8239
True Positive Rate (TPR)	1.0000	Recovered NC	0.9977
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

Experiment 13: Mandrill — Attack: Crop 33%

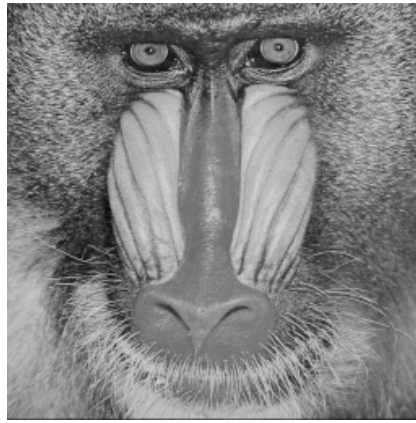


Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	32.65 dB
Watermarked SSIM	0.9989	Recovered SSIM	0.9316
True Positive Rate (TPR)	1.0000	Recovered NC	0.9990
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

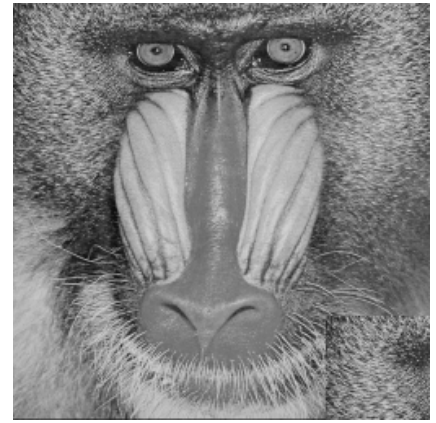
Experiment 14: Mandrill — Attack: Copy-Move



1. Original



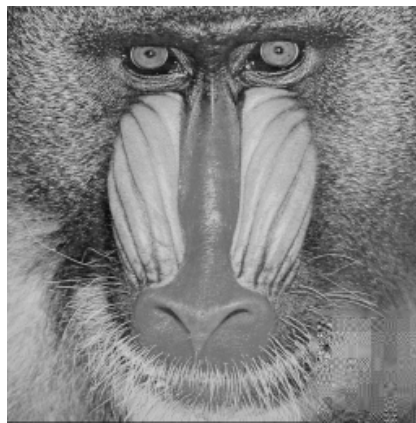
2. Watermarked



3. Attacked



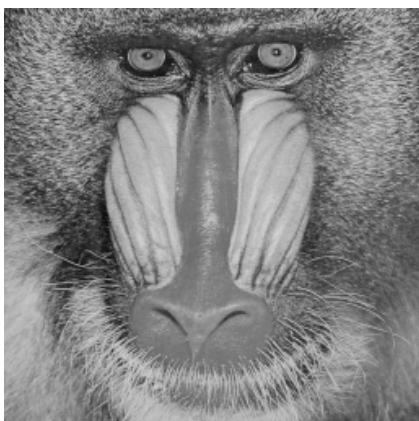
4. True Tamper Map



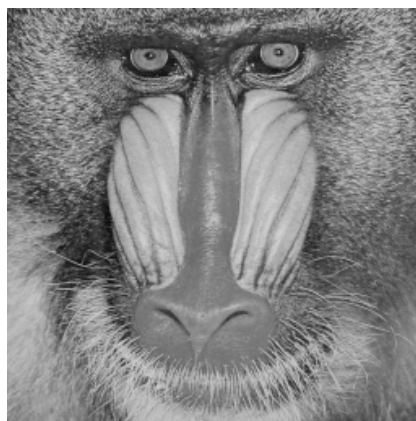
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	34.56 dB
Watermarked SSIM	0.9989	Recovered SSIM	0.9609
True Positive Rate (TPR)	1.0000	Recovered NC	0.9994
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

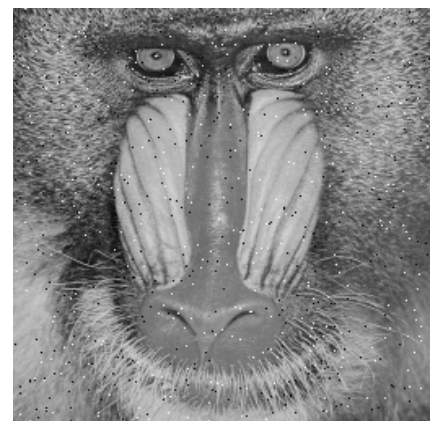
Experiment 15: Mandrill — Attack: Salt_and_Pepper_1%



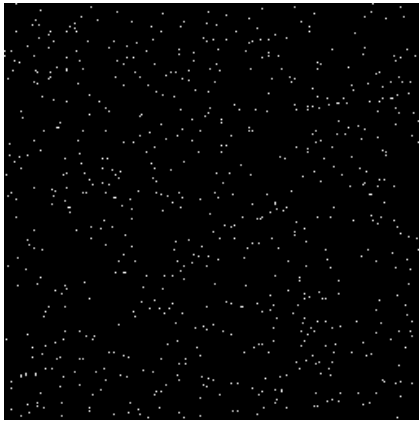
1. Original



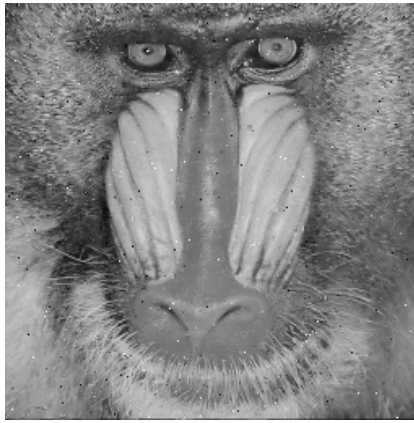
2. Watermarked



3. Attacked



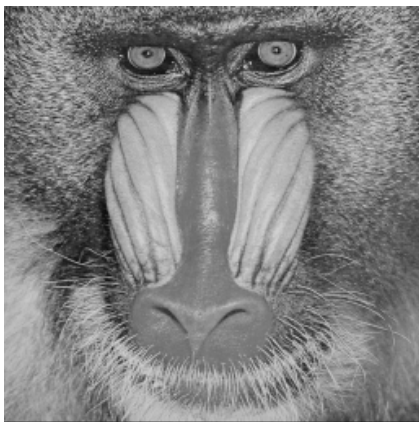
4. True Tamper Map



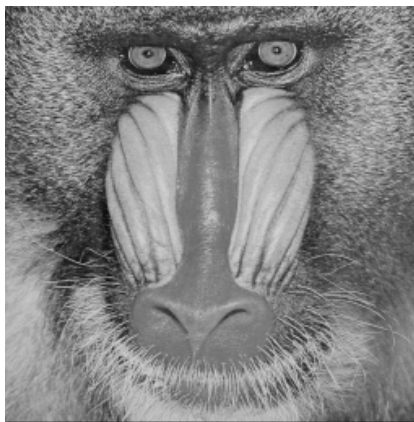
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	27.21 dB
Watermarked SSIM	0.9989	Recovered SSIM	0.8662
True Positive Rate (TPR)	0.6050	Recovered NC	0.9966
False Positive Rate (FPR)	0.0482	Tamper F1-Score	0.1943

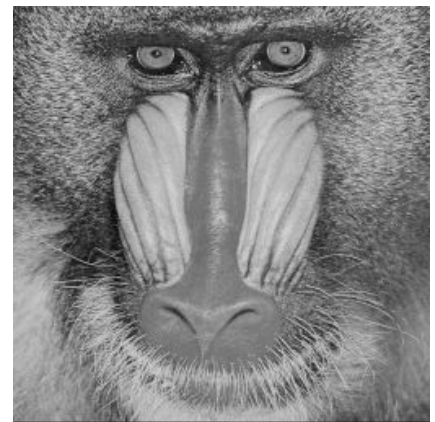
Experiment 16: Mandrill — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.15 dB	Recovered PSNR	20.26 dB
Watermarked SSIM	0.9989	Recovered SSIM	0.4345
True Positive Rate (TPR)	1.0000	Recovered NC	0.9832
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

Experiment 17: Walter Cronkite — Attack: Crop 33%



1. Original

2. Watermarked

3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.10 dB	Recovered PSNR	38.96 dB
Watermarked SSIM	0.9953	Recovered SSIM	0.9712
True Positive Rate (TPR)	1.0000	Recovered NC	0.9996
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 18: Walter Cronkite — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.10 dB	Recovered PSNR	48.20 dB
Watermarked SSIM	0.9953	Recovered SSIM	0.9933
True Positive Rate (TPR)	1.0000	Recovered NC	1.0000
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 19: Walter Cronkite — Attack: Salt_and_Pepper_1%



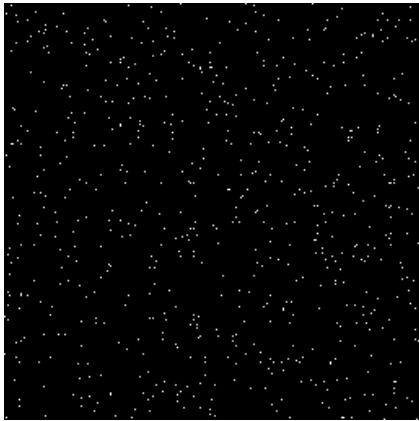
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.10 dB	Recovered PSNR	29.36 dB
Watermarked SSIM	0.9953	Recovered SSIM	0.8880
True Positive Rate (TPR)	0.5760	Recovered NC	0.9964
False Positive Rate (FPR)	0.0461	Tamper F1-Score	0.1936

Experiment 20: Walter Cronkite — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



5. Recovered Image

4. True Tamper Map

Metric	Value	Metric	Value
Watermarked PSNR	51.10 dB	Recovered PSNR	32.49 dB
Watermarked SSIM	0.9953	Recovered SSIM	0.8691
True Positive Rate (TPR)	0.9999	Recovered NC	0.9983
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

Experiment 21: Chemical plant — Attack: Crop 33%



1. Original

2. Watermarked

3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	30.79 dB
Watermarked SSIM	0.9986	Recovered SSIM	0.9382
True Positive Rate (TPR)	1.0000	Recovered NC	0.9979
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 22: Chemical plant — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	33.33 dB
Watermarked SSIM	0.9986	Recovered SSIM	0.9749
True Positive Rate (TPR)	1.0000	Recovered NC	0.9988
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 23: Chemical plant — Attack: Salt_and_Pepper_1%



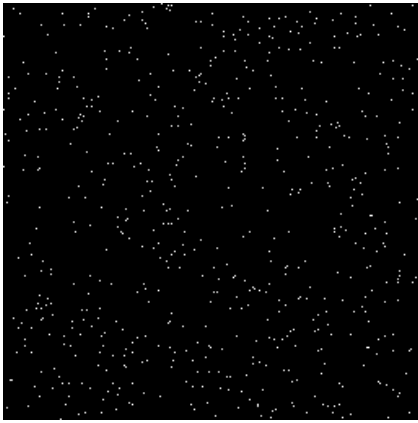
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	28.32 dB
Watermarked SSIM	0.9986	Recovered SSIM	0.8907
True Positive Rate (TPR)	0.5832	Recovered NC	0.9962
False Positive Rate (FPR)	0.0425	Tamper F1-Score	0.1933

Experiment 24: Chemical plant — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.13 dB	Recovered PSNR	22.47 dB
Watermarked SSIM	0.9986	Recovered SSIM	0.5734
True Positive Rate (TPR)	1.0000	Recovered NC	0.9855
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

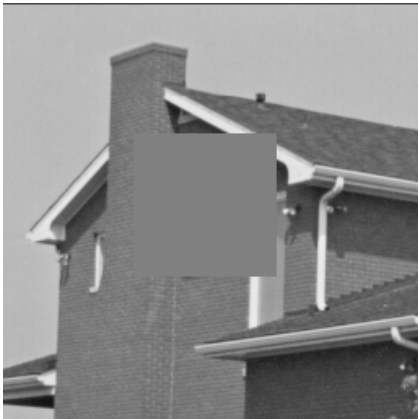
Experiment 25: House — Attack: Crop 33%



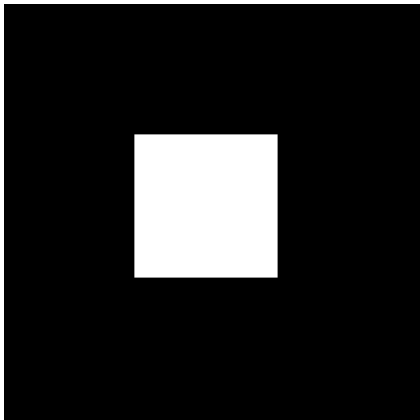
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	35.70 dB
Watermarked SSIM	0.9958	Recovered SSIM	0.9508
True Positive Rate (TPR)	1.0000	Recovered NC	0.9996
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

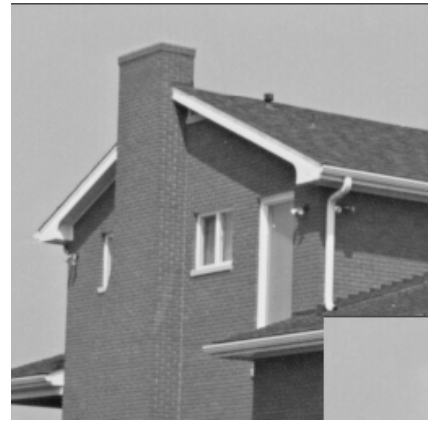
Experiment 26: House — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	37.23 dB
Watermarked SSIM	0.9958	Recovered SSIM	0.9759
True Positive Rate (TPR)	1.0000	Recovered NC	0.9997
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 27: House — Attack: Salt_and_Pepper_1%



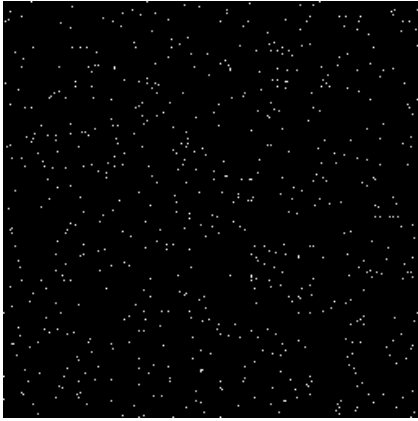
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	29.57 dB
Watermarked SSIM	0.9958	Recovered SSIM	0.8735
True Positive Rate (TPR)	0.5696	Recovered NC	0.9983
False Positive Rate (FPR)	0.0424	Tamper F1-Score	0.1923

Experiment 28: House — Attack: JPEG_80



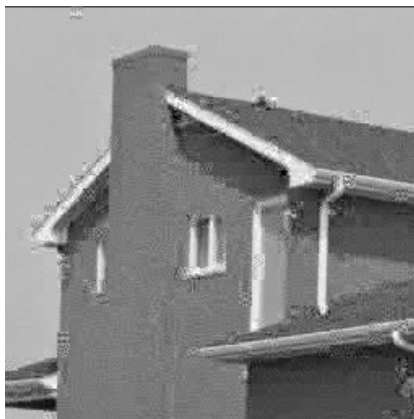
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.17 dB	Recovered PSNR	28.13 dB
Watermarked SSIM	0.9958	Recovered SSIM	0.7614
True Positive Rate (TPR)	1.0000	Recovered NC	0.9976
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000

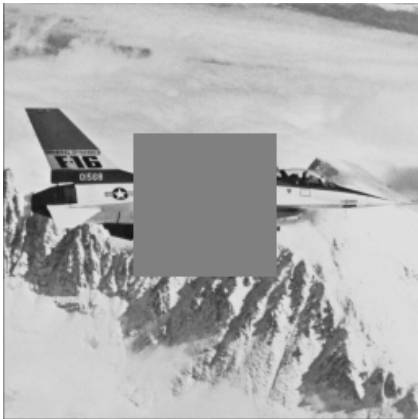
Experiment 29: Airplane — Attack: Crop 33%



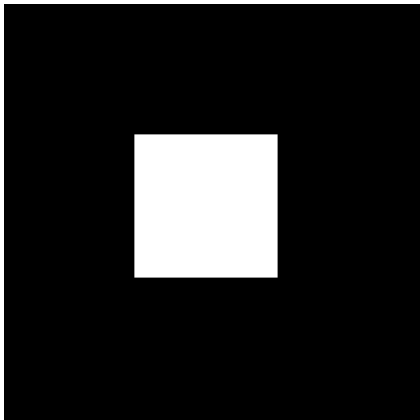
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.11 dB	Recovered PSNR	30.49 dB
Watermarked SSIM	0.9963	Recovered SSIM	0.9570
True Positive Rate (TPR)	1.0000	Recovered NC	0.9992
False Positive Rate (FPR)	0.0058	Tamper F1-Score	0.9788

Experiment 30: Airplane — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.11 dB	Recovered PSNR	33.69 dB
Watermarked SSIM	0.9963	Recovered SSIM	0.9743
True Positive Rate (TPR)	1.0000	Recovered NC	0.9996
False Positive Rate (FPR)	0.0018	Tamper F1-Score	0.9865

Experiment 31: Airplane — Attack: Salt_and_Pepper_1%



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.11 dB	Recovered PSNR	27.66 dB
Watermarked SSIM	0.9963	Recovered SSIM	0.8801
True Positive Rate (TPR)	0.5660	Recovered NC	0.9984
False Positive Rate (FPR)	0.0465	Tamper F1-Score	0.1905

Experiment 32: Airplane — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	51.11 dB	Recovered PSNR	22.99 dB
Watermarked SSIM	0.9963	Recovered SSIM	0.7027
True Positive Rate (TPR)	1.0000	Recovered NC	0.9952
False Positive Rate (FPR)	0.0000	Tamper F1-Score	1.0000
